



Cybersecurity Piscine

ft_otp

Summary: Nothing ever lasts forever...

Version: 1.00

Chapter I

Introduction

Passwords are one of the biggest headaches in computer security. Users forget them, share them, reuse them and choose them horribly bad. Furthermore, passwords are sooner or later leaked in security breaches. One way to avoid this is to use one-time passwords, based on timestamps, which expire after a few minutes and then become invalid. Whether you already use this system, or if you have never heard of it, it is quite likely that one of your passwords has been compromised at some point in your life.

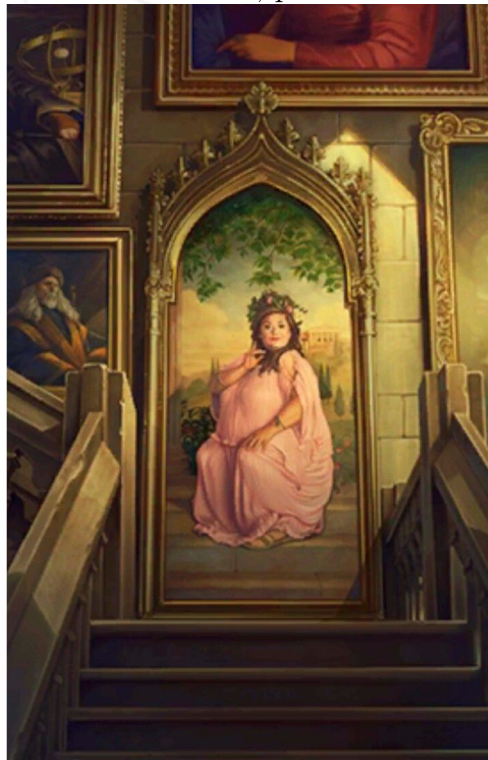
In this project, the aim is to implement a TOTP (Time-based One-Time Password) system, which will be capable of generating ephemeral passwords from a master key. It will be based on the RFC: <https://datatracker.ietf.org/doc/html/rfc6238>, so you could use it in your day to day.

Chapter II

Prologue

The Silk Road stretched across the entire Asian continent, connecting China with Mongolia, Persia, India, the Middle East, Turkey, Europe, and Africa. Despite the name, it was not the valuable cloth that was primarily traded. Glass, leather, weapons or war machines traveled the world, expanding industrial discoveries, printing techniques, gunpowder or the compass.

Password, please?



Chapter III

Mandatory Part

In the language of your choice, you have to implement a program that allows you to store an initial password in file, and that is capable of generating a new one time password every time it is requested.

You can use any library that facilitates the implementation of the algorithm, as long as it doesn't do the dirty work, i.e. using a TOTP library is strictly prohibited. Of course, you can and should use a library or function that allows you to access system time.

- The executable must be named `ft_otp`
- Your program must take arguments.
 - `-g`: The program receives as argument a hexadecimal key of at least 64 characters. The program stores this key safely in a file called `ft_otp.key`, which is encrypted.
 - `-k`: The program generates a new temporary password based on the key given as argument and prints it on the standard output.
- Your program must use the HOTP algorithm (RFC 4226).
- The generated one-time password must be random and must always contain the same format, i.e. 6 digits.

Below is an example of use:

```
$ echo -n "NEVER GONNA GIVE YOU UP" > key.txt
$ ./ft_otp -g key.txt
./ft_otp: error: key must be 64 hexadecimal characters.
$ [...]
$ cat key.hex | wc -c
64
$ ./ft_otp -g key.hex
Key was successfully saved in ft_otp.key.
$ ./ft_otp -k ft_otp.key
836492
$ sleep 60
$ ./ft_otp -k ft_otp.key
123518
```

You can check if your program is working properly by comparing generated passwords with `0athtool` or any tool of your choice.



```
oathtool -totp $(cat key.hex)
```



Please note that the reference software you choose will be used during your evaluation.

Chapter V

Submission and peer-evaluation

Turn in your assignment in your `Git` repository as usual. Only the work inside your repository will be evaluated during the defense. Don't hesitate to double check the names of your folders and files to ensure they are correct.